The Derivative of $|x|$ from First Principles

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1 Setup

Let $f(x) = |x|$. For fixed x and $h \neq 0$, define the difference quotient

$$Q_x(h) = \frac{|x+h| - |x|}{h}.$$

By definition, $f'(x) = \lim_{h\to 0} Q_x(h)$, provided the limit exists.

Definition 1.1. Let g be defined on a punctured neighborhood of 0. Then $\lim_{h\to 0} g(h) = L$ means: for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$0 < |h| < \delta \implies |g(h) - L| < \varepsilon.$$

Two features of this definition are used throughout.

1. The number δ is chosen by the prover. If some δ works, every smaller positive δ also works, since it only restricts h further.
2. The condition says nothing about h with $|h| \geq \delta$. Values of g outside the chosen neighborhood never enter the definition. A limit is a local property.

Lemma 1.2 (Local agreement)**.** *Suppose $g(h) = k(h)$ for all h with $0 < |h| < r$, for some $r > 0$. Then $\lim_{h\to 0} g(h) = L$ if and only if $\lim_{h\to 0} k(h) = L$.*

Proof. Suppose $\lim_{h\to 0} k(h) = L$. Given $\varepsilon > 0$, let δ be the number provided for k, and set $\delta' = \min(\delta, r)$. If $0 < |h| < \delta'$, then $g(h) = k(h)$ and $|k(h) - L| < \varepsilon$, so $|g(h) - L| < \varepsilon$. The converse is symmetric. □

Every derivation below reduces the problem to this lemma: one shows that $Q_x(h)$ agrees with a simpler expression for all sufficiently small $h \neq 0$, and then takes the limit of the simpler expression.

2 Method 1: Case analysis

The absolute value is defined piecewise: $|a| = a$ for $a \geq 0$ and $|a| = -a$ for $a \leq 0$. The task is to show that, for h small enough, $x + h$ has the same sign as x, so that a single piece of the definition applies to both $|x + h|$ and $|x|$.

2.1 The case $x > 0$

Proposition 2.1. *If $x > 0$, then $f'(x) = 1$.*

Proof. Take $0 < |h| < x$. Since $h \geq -|h|$, with equality exactly when $h < 0$,

$$x + h \geq x - |h| > 0.$$

The first inequality cannot be made strict: for $h < 0$ it is an equality. Strictness of the conclusion comes from the second inequality, which follows from $|h| < x$; a chain $a \geq b > c$ gives $a > c$.

Hence $|x + h| = x + h$ and $|x| = x$, so

$$Q_x(h) = \frac{(x + h) - x}{h} = \frac{h}{h} = 1.$$

Given any $\varepsilon > 0$, choose $\delta = x$. Then $0 < |h| < \delta$ implies $|Q_x(h) - 1| = 0 < \varepsilon$. Thus $f'(x) = 1$. □

Here a single δ works for every ε. This happens only because Q_x is exactly constant near 0.

2.2 The case $x < 0$

Proposition 2.2. *If $x < 0$, then $f'(x) = -1$.*

Proof. Take $0 < |h| < |x| = -x$. Since $h \leq |h|$, with equality exactly when $h > 0$,

$$x + h \leq x + |h| < 0.$$

Hence $|x + h| = -(x + h)$ and $|x| = -x$, so

$$Q_x(h) = \frac{-(x + h) - (-x)}{h} = \frac{-h}{h} = -1.$$

Choosing $\delta = -x$ completes the proof as before. □

2.3 The case $x = 0$

For $h \neq 0$,

$$Q_0(h) = \frac{|h| - 0}{h} = \frac{|h|}{h} = \begin{cases} 1, & h > 0, \\ -1, & h < 0. \end{cases}$$

The right-hand limit is 1 and the left-hand limit is -1.

Proposition 2.3. *f is not differentiable at 0.*

Proof. Suppose $\lim_{h \to 0} Q_0(h) = L$ for some real L. Take $\varepsilon = 1$ and let $\delta > 0$ be the corresponding number. Both $h_1 = \delta/2$ and $h_2 = -\delta/2$ satisfy $0 < |h| < \delta$. From h_1: $|1 - L| < 1$, so $0 < L < 2$. From h_2: $|-1 - L| < 1$, so $-2 < L < 0$. These are incompatible. □

2.4 Remarks on the choice of neighborhood

Remark 2.4 (Why not arbitrary h). The identity $Q_x(h) = 1$ for $x > 0$ is false for large negative h. For example, with $x = 1$ and $h = -3$,

$$Q_1(-3) = \frac{|-2| - |1|}{-3} = \frac{2-1}{-3} = -\frac{1}{3}.$$

Once $h < -x$, the sign of $x + h$ flips and the formula $|x + h| = x + h$ no longer applies. This does not affect the limit: by feature 2 of the definition and Lemma 1.2, only the values of Q_x on some punctured neighborhood of 0 matter, and the restriction $0 < |h| < x$ is precisely the choice $\delta = x$ that the prover is entitled to make.

Remark 2.5 (Non-strict neighborhood). One may instead take $0 < |h| \leq x$. Then $h = -x$ is allowed and gives $x + h = 0$, so the claim $x + h > 0$ fails. What survives is

$$x + h \geq x - |h| \geq 0,$$

which suffices, because $|a| = a$ holds for all $a \geq 0$. Directly,

$$Q_x(-x) = \frac{0 - x}{-x} = 1.$$

Similarly for $x < 0$ with $0 < |h| \leq -x$: at $h = -x$ we have $x + h = 0$, and $|a| = -a$ holds for all $a \leq 0$, so $Q_x(-x) = -1$.

Nothing is gained. Any smaller neighborhood, such as $0 < |h| < x/2$, works equally well. The strict version is conventional because it matches the form $0 < |h| < \delta$ in the definition and avoids the boundary case $x + h = 0$.

Remark 2.6 (The largest set). For $x > 0$, the set of all $h \neq 0$ with $Q_x(h) = 1$ is not symmetric: it is $\{h \neq 0 : h \geq -x\}$, since positive h of any size keeps $x + h > 0$. The symmetric neighborhood is chosen only because it fits the definition of the limit directly.

3 Method 2: Rationalization

3.1 Motivation

The identity $|a| = \sqrt{a^2}$ rewrites the difference quotient as

$$Q_x(h) = \frac{\sqrt{(x+h)^2} - \sqrt{x^2}}{h},$$

which has the structure of the first-principles problem for $\sqrt{x}$: a difference of two roots over h. The standard technique there is to multiply numerator and denominator by the sum of the roots, so that the identity $(A - B)(A + B) = A^2 - B^2$ removes the roots. The same technique applies with $|\cdot|$ in place of $\sqrt{\cdot}$.

The quotient as written has the form $0/0$ in the limit and cannot be evaluated by substituting $h = 0$. The goal is to rewrite it until a factor of h cancels.

3.2 Derivation

Fix $x \neq 0$ and $h \neq 0$. Write $A = |x + h|$ and $B = |x|$.

Step 1: Multiply by 1 in a useful form.

$$\frac{A-B}{h} = \frac{A-B}{h} \cdot \frac{A+B}{A+B}.$$

This requires $A + B \neq 0$. Since $B = |x| > 0$ and $A \geq 0$, we have $A + B \geq |x| > 0$. This is the first use of $x \neq 0$.

Step 2: Difference of squares. Since $(A - B)(A + B) = A^2 - B^2$,

$$\frac{A-B}{h} = \frac{A^2 - B^2}{h\,(A+B)}.$$

Step 3: Remove the absolute values in the numerator. For every real a, $|a|^2 = a^2$. Hence

$$A^2 = (x+h)^2, \qquad B^2 = x^2.$$

This is the essential step. Before it, the numerator contains absolute values, which force a case split. After it, the numerator is a polynomial. The absolute values remain in the denominator, where they cause no difficulty.

Step 4: Expand.

$$(x+h)^2 - x^2 = (x^2 + 2xh + h^2) - x^2 = 2xh + h^2 = h(2x+h).$$

Step 5: Cancel h.

$$Q_x(h) = \frac{h(2x+h)}{h\,(|x+h| + |x|)} = \frac{2x+h}{|x+h| + |x|}.$$

The cancellation is valid because $h \neq 0$. The two expressions agree for all $h \neq 0$, which is a punctured neighborhood of 0, so by Lemma 1.2 they have the same limit.

Step 6: Take the limit. The numerator satisfies $2x + h \to 2x$. For the denominator, the reverse triangle inequality gives

$$\big|\, |x+h| - |x| \,\big| \leq |h| \to 0,$$

so $|x + h| \to |x|$ and $|x + h| + |x| \to 2|x|$. The quotient rule for limits requires the limit of the denominator to be nonzero; it is $2|x| > 0$, which is the second use of $x \neq 0$. Therefore

$$f'(x) = \lim_{h \to 0} \frac{2x+h}{|x+h| + |x|} = \frac{2x}{2|x|} = \frac{x}{|x|}.$$

Step 7: Read off the sign. $x/|x|$ equals 1 for $x > 0$ and -1 for $x < 0$, in agreement with Method 1.

3.3 Where each hypothesis is used

Step	Requirement	Reason
1	$\|x+h\| + \|x\| \neq 0$	multiplying by $(A+B)/(A+B)$
3	none	$\|a\|^2 = a^2$ for all real a
5	$h \neq 0$	cancelling h; always true inside a limit
6	$\|x\| \neq 0$	quotient rule needs a nonzero denominator limit

3.4 What happens at $x = 0$

With $x = 0$, Step 1 requires $|h| \neq 0$, which holds. Steps 2–5 give

$$Q_0(h) = \frac{2 \cdot 0 + h}{|h| + 0} = \frac{h}{|h|}.$$

Step 6 fails: the denominator tends to 0, the quotient rule does not apply, and $h/|h| = \pm 1$ according to the sign of h. This is the original quotient $|h|/h$ again, which has no limit. The algebra locates the failure exactly.

4 Method 3: The chain rule

This method is not from first principles, since it presupposes the derivative of $\sqrt{u}$, but it shows the same structure from another angle.

Write $f(x) = \sqrt{x^2}$, the composition of $x \mapsto x^2$ with $u \mapsto \sqrt{u}$. For $x \neq 0$, the outer function is differentiable at $u = x^2 > 0$ with derivative $1/(2\sqrt{u})$. The chain rule gives

$$f'(x) = \frac{1}{2\sqrt{x^2}} \cdot 2x = \frac{x}{|x|}.$$

The derivative of $\sqrt{u}$ for $u > 0$ is itself obtained by rationalization, so this method is Method 2 packaged through a theorem.

At $x = 0$ the outer function is evaluated at $u = 0$, where $\sqrt{u}$ is not differentiable, so the chain rule does not apply. This proves nothing about $f'(0)$: the chain rule is a sufficient condition, and the failure of a sufficient condition does not establish non-differentiability. For example, $x^2 = \sqrt{x^4}$ has the same form, with $\sqrt{u}$ again evaluated at $u = 0$ when $x = 0$, yet x^2 is differentiable at 0. Non-differentiability of $|x|$ at 0 must be shown directly, as in Method 1.

5 Method 4: Locality of the derivative

Method 1 can be stated as a general principle.

Proposition 5.1. *Suppose $f(t) = g(t)$ for all t in an open interval $(x - r, x + r)$, and g is differentiable at x. Then f is differentiable at x and $f'(x) = g'(x)$.*

Proof. For $0 < |h| < r$, both x and $x + h$ lie in the interval, so

$$\frac{f(x+h) - f(x)}{h} = \frac{g(x+h) - g(x)}{h}.$$

The difference quotients agree on a punctured neighborhood of 0, and Lemma 1.2 gives the result. □

For $x > 0$, $|t| = t$ on $(0, 2x) = (x - x, x + x)$, so $f'(x) = \frac{d}{dt}t = 1$. For $x < 0$, $|t| = -t$ on $(2x, 0)$, so $f'(x) = -1$. The radius $r = |x|$ is the same restriction $0 < |h| < |x|$ used in Method 1.

At $x = 0$ no open interval around 0 lies within a single piece. As with the chain rule, this observation alone does not prove non-differentiability. The one-sided version of the proposition does apply: $|t| = t$ on $[0, \infty)$ and $|t| = -t$ on $(-\infty, 0]$, so the right derivative at 0 is 1 and the left derivative is -1. Since they differ, the two-sided derivative does not exist.

6 Result

$$\frac{d}{dx}|x| = \frac{x}{|x|} = \operatorname{sgn}(x), \qquad x \neq 0,$$

and the derivative does not exist at $x = 0$. The function $|x|$ is continuous at 0 but not differentiable there.

Remark 6.1. The one-sided derivatives at 0 exist:

$$f'_+(0) = \lim_{h \to 0^+} \frac{|h|}{h} = 1, \qquad f'_-(0) = \lim_{h \to 0^-} \frac{|h|}{h} = -1.$$

Remark 6.2. In convex analysis the subdifferential of $|x|$ at 0 is the interval $[-1, 1]$, the set of slopes of lines through the origin lying below the graph. This is a different object from the derivative.

Remark 6.3. The formula $\frac{d}{dx}|x| = \operatorname{sgn}(x)$ holds only on $x \neq 0$. It is an error to extend it to 0 by the convention $\operatorname{sgn}(0) = 0$: the derivative at 0 does not exist, so it has no value, 0 or otherwise.

7 Self-check questions

Q1. Why may the derivation for $x > 0$ restrict to $0 < |h| < x$?

Answer. The definition of the limit lets the prover choose δ, and only values with $0 < |h| < \delta$ are examined. Choosing $\delta = x$ is legitimate; values of Q_x outside this neighborhood are irrelevant.

Q2. Compute $Q_1(-3)$. What does the result show?

Answer. $Q_1(-3) = -1/3$. The identity $Q_x(h) = 1$ fails for large negative h, which is why the neighborhood must be restricted.

Q3. Does the derivation for $x > 0$ still work with $0 < |h| \leq x$?

Answer. Yes. At $h = -x$, $x + h = 0$ and $|0| = 0 = x + h$, so $Q_x(-x) = 1$. The justification uses $x + h \geq 0$ instead of $x + h > 0$.

Q4. In the rationalization method, which step removes the absolute values, and why does it work?

Answer. Step 3, using $|a|^2 = a^2$ for all real a. Squaring discards the sign, turning the numerator into a polynomial.

Q5. Where is $x \neq 0$ used in the rationalization method?

Answer. In Step 1, to guarantee $|x + h| + |x| \neq 0$, and in Step 6, so that the denominator tends to $2|x| \neq 0$ and the quotient rule for limits applies.

Q6. Why is cancelling h in Step 5 valid?

Answer. In the limit $h \to 0$, h is never 0. The two expressions agree on the punctured neighborhood $h \neq 0$, so by local agreement they have the same limit.

Q7. The chain rule does not apply to $\sqrt{x^2}$ at $x = 0$. Does this prove $|x|$ is not differentiable at 0?

Answer. No. The chain rule is a sufficient condition. The function $\sqrt{x^4} = x^2$ has the same obstruction and is differentiable at 0. Non-differentiability requires the direct one-sided argument.

Q8. Is $\frac{d}{dx}|x|$ equal to 0 at $x = 0$?

Answer. No. The derivative does not exist at 0. The one-sided derivatives are 1 and -1.